

Assignment 1 & 2, 2024

Level: BE

Subject : Algebra and Geometry

Due Date : 20th may , 2025

Attempt all questions.

1. Solve the following system of linear equations by Cramer's rule

$$2x - y + 3z = 9, x + y + z = 6, x - y + z = 2$$

2. Find the rank of matrix $\begin{pmatrix} 2 & -2 & 0 & 6 \\ 4 & 2 & 0 & 2 \\ 1 & -1 & 0 & 3 \\ 1 & -2 & 1 & 2 \end{pmatrix}$

3. Test the consistency and then find the solution if exists by using Gauss elimination method

(a) $x + 2y - 3z = 9, 2x - y + 2z = -8, 3x - y - 4z = 3$

(b) $x + y - 2z = -2, x - 2y + z = 1, -2x + y + z = 1$

(c) $x - 2y + z = 1, x + y - 2z = 1, -2x + y + z = 1$

4. Determine the value of p and q for which the system of equations

$$x - 2y + 2z = 2, x + 2y + 3z = -1, 2x + 2y + pz = q \text{ has}$$

- a. a unique solution
- b. no solution
- c. infinitely many solutions

5. Let $V = \mathbb{R}^3$ be a vector space. Then show that $W = \{(x, y, 0) : x, y \in \mathbb{R}\}$ is a vector subspace of V .

6. Let $V = \mathbb{R}^3$ be a vector space over a field $\mathbb{R}$. Then show that $W = \{(x, y, z) : x + y + z = 0\}$ is a subspace of V .

7. Let V be the vector space over a field $\mathbb{R}$ of all 2×2 matrices. Then show that

$$W = \left\{ \begin{pmatrix} a & b \\ c & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\} \text{ is a subspace of } V.$$

8. Check whether the vectors $(1, 2), (2, 0)$ form a basis of $\mathbb{R}^2$ or not.

9. Check whether the vectors $(1, 2, 0), (0, 1, 1), (1, 0, 3)$ form a basis of $\mathbb{R}^3$ or not.

10. Check whether the vectors $(1, 2, -1), (2, 1, 0), (5, 4, 0)$ form a basis of $\mathbb{R}^3$ or not.

11. Determine whether the following transformation is linear or not?

a. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x, x + y)$

b. $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x + y - z, 0)$

c. $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x - 2y, y + 2z, x + 3z)$

d. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x, xy)$

12. Find the eigen values and eigen vectors of the following matrices.

a. $\begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$ b. $\begin{pmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{pmatrix}$ c. $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ d. $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

13. Verify Cayley Hamilton Theorem for the following matrices. Also, find the inverse of the matrices using this theorem.

a. $\begin{pmatrix} 1 & 2 \\ 3 & -5 \end{pmatrix}$ b. $\begin{pmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{pmatrix}$

14. Diagonalize the matrix $\begin{pmatrix} 2 & 1 \\ 5 & 6 \end{pmatrix}$

15. Maximize $Z = 3x_1 + 2x_2$ under the constraints

$$x_1 + x_2 \leq 4$$

$$x_1 - x_2 \leq 2$$

$$x_1, x_2 \geq 0.$$

16. Maximize $Z = 30x_1 + 20x_2$ under the constraints

$$-x_1 + x_2 \leq 5$$

$$2x_1 + x_2 \leq 10$$

$$x_1, x_2 \geq 0.$$

17. Maximize $Z = 3x_1 + 5x_2 + 4x_3$ under the constraints

$$2x_1 + 3x_3 \leq 8$$

$$5x_1 + 2x_2 + 2x_3 \leq 10$$

$$5x_2 + 4x_3 \leq 15$$

$$x_1, x_2, x_3 \geq 0.$$

18. Minimize $Z = 2x_1 - 10x_2$ under the constraints

$$x_1 - x_2 \leq 4$$

$$2x_1 + x_2 \leq 14$$

$$x_1 + x_2 \leq 9$$

$$-x_1 + 3x_2 \leq 15$$

$$x_1, x_2 \geq 0.$$

19. Construct the dual problem of the following primal problem and solve by using simplex method.

Minimize $Z = 3x_1 + 2x_2$ under the constraints

$$7x_1 + 2x_2 \geq 30$$

$$5x_1 + 4x_2 \geq 20$$

$$2x_1 + 8x_2 \geq 16$$

$$x_1, x_2 \geq 0.$$

20. Solve the following LPP by Big M method

Maximize $Z = -3x_1 + 7x_2$ under the constraints

$$2x_1 + 3x_2 \leq 5$$

$$5x_1 + 2x_2 \geq 3$$

$$x_2 \leq 1$$

$$x_1, x_2 \geq 0.$$

21. Solve the following LPP by Big M method

Maximize $Z = x_1 - x_2 + 3x_3$ under the constraints

$$x_1 + x_2 \leq 20$$

$$x_1 + x_3 = 5$$

$$x_2 + x_3 \geq 10$$

$$x_1, x_2, x_3 \geq 0.$$

22. One unit of product A contributes Rs 7 and requires 3 units of raw materials and 2 hrs of labour.

One unit of product B contributes Rs 5 and requires one unit of raw material and one hour labour.

Availability of raw materials at present is 48 units and there are 40 hours of labour.

a. Formulate this problem as a LPP

b. Write its dual and then find the optimal solution (maximum profit).

23. A company produces three products P, Q and R from three raw materials A, B and C. One unit of P requires 2 units of A and 3 of B. Product Q requires 2 units of B and 5 of C. One unit of R requires 3 units of A, 2 units of B and 4 units of C. Company has 8 units of material A, 10 units of material B and 15 units of material C available to it. Profit per unit of product P, Q and are Rs 3, Rs 5 and Rs. 4 respectively. How many units of each product should be produced to maximize profit?

"ALL THE BEST."